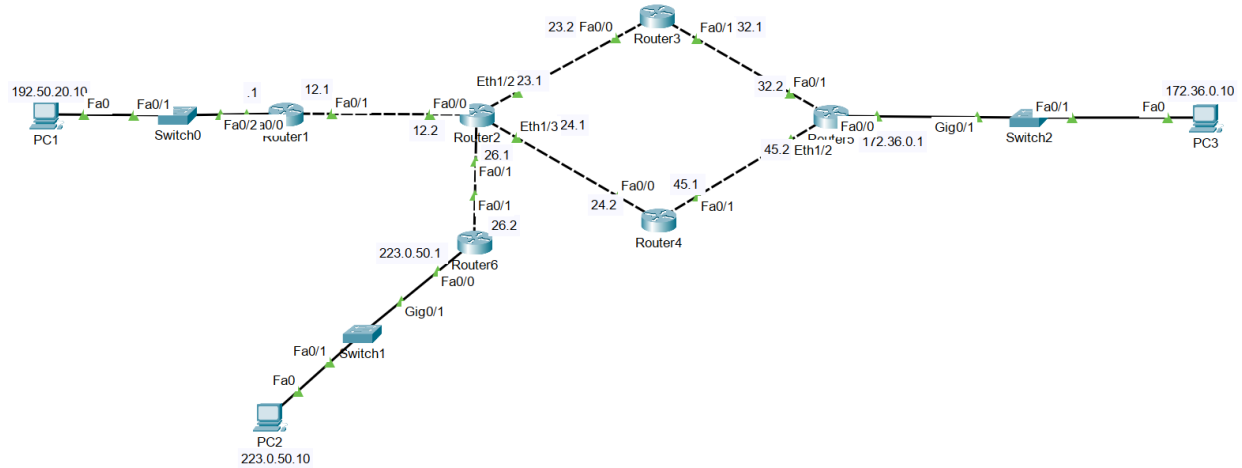


Static Route Project-2



Ip configuration In PCs

PC1	PC2	PC3
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\> C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix...: Link-local IPv6 Address...: FE80::207:ECFF IPv6 Address...: IPv4 Address...: 192.50.20.10 Subnet Mask...: 255.255.255.0 Default Gateway...: 192.50.20.1 Bluetooth Connection: Connection-specific DNS Suffix...: Link-local IPv6 Address...: IPv6 Address...: IPv4 Address...: 0.0.0.0 Subnet Mask...: 0.0.0.0 Default Gateway...: 0.0.0.0</pre>	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\> ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix...: Link-local IPv6 Address...: FE80::209:7CFF IPv6 Address...: IPv4 Address...: 223.0.50.10 Subnet Mask...: 255.255.255.0 Default Gateway...: 223.0.50.1 Bluetooth Connection: Connection-specific DNS Suffix...: Link-local IPv6 Address...: IPv6 Address...: IPv4 Address...: 0.0.0.0 Subnet Mask...: 0.0.0.0 Default Gateway...: 0.0.0.0 C:\></pre>	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\> ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix...: Link-local IPv6 Address...: FE80::260:2FF IPv6 Address...: IPv4 Address...: 172.36.0.10 Subnet Mask...: 255.255.0.0 Default Gateway...: Bluetooth Connection: Connection-specific DNS Suffix...: Link-local IPv6 Address...: IPv6 Address...: IPv4 Address...: 0.0.0.0 Subnet Mask...: 0.0.0.0 Default Gateway...: 0.0.0.0 C:\></pre>

Configure Ip for Router Interfaces

Cisco Packet Tracer - C:\Users\Ansath\Desktop\CCNA SUREN NETWORK\Lab5\static route project 2.pkt

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.50.20.10 255.255.255.0
Router(config-if)#ip address 192.50.20.10 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 12.0.0.1 255.0.0.0
Router(config-if)#ip address 12.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/0    192.50.20.10    YES manual up
up
FastEthernet0/1    12.0.0.1        YES manual up
down
```

Router6

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 223.0.50.1 255.255.255.0
Router(config-if)#ip address 223.0.50.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to u
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/
state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 26.0.0.2 255.0.0.0
Router(config-if)#ip address 26.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to u
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/
state to up

Router(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/0    223.0.50.1      YES manual up
up
FastEthernet0/1    26.0.0.2        YES manual up
down
```

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 23.0.0.2 255.0.0.0
Router(config-if)#ip address 23.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 32.0.0.1 255.0.0.0
Router(config-if)#ip address 32.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/0    23.0.0.2        YES manual up
up
FastEthernet0/1    32.0.0.1        YES manual up
down
Router(config-if)#
```

Router5

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 172.36.0.1 255.255.0.0
Router(config-if)#ip address 172.36.0.1 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
ip address 32.0.0.2 255.0.0.0
Router(config-if)#ip address 32.0.0.2 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet1/2
Router(config-if)#ip address 45.0.0.2 255.0.0.0
Router(config-if)#ip address 45.0.0.2 255.0.0.0
Router(config-if)#no shutdown
```

Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 45.0.0.1 255.0.0.0
Router(config-if)#ip address 45.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/0    24.0.0.2        YES manual up
up
FastEthernet0/1    45.0.0.1        YES manual up
down
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
```

Configure Static Routes

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#ip route 26.0.0.0 255.0.0.0 12.0.0.2
Router(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2
Router(config)#ip route 23.0.0.0 255.0.0.0 12.0.0.2
Router(config)#ip route 24.0.0.0 255.0.0.0 12.0.0.2
Router(config)#ip route 32.0.0.0 255.0.0.0 12.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 12.0.0.2
Router(config)#ip route 172.36.0.10 255.255.0.0 12.0.0.2
%Inconsistent address and mask
Router(config)#ip route 172.36.0.0 255.255.0.0 12.0.0.2
Router(config)#
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
        area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
        2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E -
        EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
        IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C 12.0.0.0/8 is directly connected, FastEthernet0/1
S 23.0.0.0/8 [1/0] via 12.0.0.2
S 24.0.0.0/8 [1/0] via 12.0.0.2
S 26.0.0.0/8 [1/0] via 12.0.0.2
S 32.0.0.0/8 [1/0] via 12.0.0.2
S 45.0.0.0/8 [1/0] via 12.0.0.2
S 172.36.0.0/16 [1/0] via 12.0.0.2
C 192.50.20.0/24 is directly connected, FastEthernet0/0
S 223.0.50.0/24 [1/0] via 12.0.0.2
Router(config)#
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.50.20.0 255.255.255.0 12.0.0.1
Router(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.1
%Invalid next hop address (it's this router)
Router(config)#ip route 223.0.50.0 255.255.255.0 26.0.0.2
Router(config)#ip route 32.0.0.0 255.0.0.0 23.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 23.0.0.2
Router(config)#ip route 45.0.0.0 255.0.0.0 24.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 24.0.0.2
Router(config)#
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
        BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
        area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
        inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C 12.0.0.0/8 is directly connected, FastEthernet0/0
C 23.0.0.0/8 is directly connected, Ethernet1/2
C 24.0.0.0/8 is directly connected, Ethernet1/3
C 26.0.0.0/8 is directly connected, FastEthernet0/1
S 32.0.0.0/8 [1/0] via 23.0.0.2
S 45.0.0.0/8 [1/0] via 24.0.0.2
S 172.36.0.0/16 [1/0] via 23.0.0.2
                        [1/0] via 24.0.0.2
S 192.50.20.0/24 [1/0] via 12.0.0.1
S 223.0.50.0/24 [1/0] via 26.0.0.2
Router(config)#
```

Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#ip route 12.0.0.0 255.0.0.0 23.0.0.1
Router(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
^
% Invalid input detected at '^' marker.

Router(config)#ip route 192.50.20.0 255.255.255.0 23.0.0.1
Router(config)#ip route 26.0.0.0 255.0.0.0 23.0.0.1
Router(config)#ip route 223.0.50.0 255.255.255.0 23.0.0.1
Router(config)#ip route 36.0.0.0 255.0.0.0 32.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 32.0.0.2
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
        BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
        area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
        2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E -
        EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
        IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 23.0.0.1
C 23.0.0.0/8 is directly connected, FastEthernet0/0
S 26.0.0.0/8 [1/0] via 23.0.0.1
C 32.0.0.0/8 is directly connected, FastEthernet0/1
S 36.0.0.0/8 [1/0] via 32.0.0.2
S 172.36.0.0/16 [1/0] via 32.0.0.2
S 192.50.20.0/24 [1/0] via 23.0.0.1
S 223.0.50.0/24 [1/0] via 23.0.0.1
Router(config)#
```

Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#ip route 12.0.0.0 255.0.0.0 24.0.0.1
Router(config)#ip route 192.50.20.0 255.255.255.0 24.0.0.1
Router(config)#ip route 26.0.0.0 255.0.0.0 24.0.0.1
Router(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
Router(config)#ip route 36.0.0.0 255.0.0.0 45.0.0.2
Router(config)#ip route 172.36.0.0 255.255.0.0 45.0.0.2
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B -
        BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
        area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
        2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E -
        EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
        IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 24.0.0.1
C 24.0.0.0/8 is directly connected, FastEthernet0/0
S 26.0.0.0/8 [1/0] via 24.0.0.1
S 36.0.0.0/8 [1/0] via 45.0.0.2
C 45.0.0.0/8 is directly connected, FastEthernet0/1
S 172.36.0.0/16 [1/0] via 45.0.0.2
S 192.50.20.0/24 [1/0] via 24.0.0.1
S 223.0.50.0/24 [1/0] via 24.0.0.1
Router(config)#
```

Router5

Physical Config CLI Attributes

IOS Command Line Interface

Router(config)#ip route 23.0.0.0 255.0.0.0 32.0.0.1
Router(config)#ip route 12.0.0.0 255.0.0.0 32.0.0.1
Router(config)#ip route 192.50.20.0 255.255.255.0 32.0.0.1
Router(config)#ip route 26.0.0.0 255.0.0.0 32.0.0.1
Router(config)#ip route 223.0.50.0 255.255.255.0 32.0.0.1
Router(config)#ip route 24.0.0.0 255.0.0.0 45.0.0.1
Router(config)#ip route 12.0.0.0 255.0.0.0 45.0.0.1
Router(config)#ip route 192.50.20.0 255.255.255.0 45.0.0.1
Router(config)#ip route 26.0.0.0 255.0.0.0 45.0.0.1
Router(config)#ip route 223.0.50.0 255.255.255.0 45.0.0.1
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 32.0.0.1
[1/0] via 45.0.0.1
S 23.0.0.0/8 [1/0] via 32.0.0.1
S 24.0.0.0/8 [1/0] via 45.0.0.1
S 26.0.0.0/8 [1/0] via 32.0.0.1
[1/0] via 45.0.0.1
C 32.0.0.0/8 is directly connected, FastEthernet0/1
C 45.0.0.0/8 is directly connected, Ethernet1/2
C 172.36.0.0/16 is directly connected, FastEthernet0/0
S 192.50.20.0/24 [1/0] via 32.0.0.1
[1/0] via 45.0.0.1
S 223.0.50.0/24 [1/0] via 32.0.0.1
[1/0] via 45.0.0.1

Router6

Physical Config CLI Attributes

IOS Command Line Interface

Router(config)#
Router(config)#ip route 12.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 192.50.20.0 255.255.255.0 26.0.0.1
Router(config)#ip route 23.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 32.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 36.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 172.36.0.0 255.255.0.0 26.0.0.1
Router(config)#ip route 24.0.0.0 255.0.0.0 26.0.0.1
Router(config)#ip route 45.0.0.0 255.0.0.0 26.0.0.1
Router(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 26.0.0.1
S 23.0.0.0/8 [1/0] via 26.0.0.1
S 24.0.0.0/8 [1/0] via 26.0.0.1
C 26.0.0.0/8 is directly connected, FastEthernet0/1
C 32.0.0.0/8 [1/0] via 26.0.0.1
S 36.0.0.0/8 [1/0] via 26.0.0.1
S 45.0.0.0/8 [1/0] via 26.0.0.1
S 172.36.0.0/16 [1/0] via 26.0.0.1
S 192.50.20.0/24 [1/0] via 26.0.0.1
C 223.0.50.0/24 is directly connected, FastEthernet0/0
Router(config)#

Connectivity Check

Ping PC1 to PC2 -Success

PC1

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125

Ping statistics for 223.0.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC2

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 192.50.20.10

Pinging 192.50.20.10 with 32 bytes of data:

Reply from 192.50.20.10: bytes=32 time<1ms TTL=125
Reply from 192.50.20.10: bytes=32 time=11ms TTL=125
Reply from 192.50.20.10: bytes=32 time<1ms TTL=125
Reply from 192.50.20.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.50.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 2ms

Ping PC2 to PC3 -Success

PC3

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=11ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=10ms TTL=124

Ping statistics for 223.0.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 5ms

PC2

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=11ms TTL=124
Reply from 172.36.0.10: bytes=32 time=1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=19ms TTL=124

Ping statistics for 172.36.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 19ms, Average = 7ms

Ping PC1 to PC3 -Success

The screenshot displays two side-by-side windows from Cisco Packet Tracer. The left window is for PC1, and the right window is for PC3. Both windows have tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active in both, showing a Command Prompt window. In the PC1 Command Prompt, the command 'C:\>ping 172.36.0.10' has been executed, resulting in four successful replies with varying times (1ms, 21ms, 1ms, 11ms) and a TTL of 124. The ping statistics show 4 packets sent, 4 received, and 0% loss, with an average round trip time of 8ms. In the PC3 Command Prompt, the command 'C:\>ping 192.50.20.10' has been executed, resulting in four successful replies with times of 1ms, 1ms, 1ms, and 10ms, all with a TTL of 124. The ping statistics show 4 packets sent, 4 received, and 0% loss, with an average round trip time of 3ms.

PC	Command	Reply 1	Reply 2	Reply 3	Reply 4	Statistics
PC1	C:\>ping 172.36.0.10	bytes=32 time<1ms TTL=124	bytes=32 time=21ms TTL=124	bytes=32 time<1ms TTL=124	bytes=32 time=11ms TTL=124	Sent=4, Received=4, Lost=0 (0% loss), Average=8ms
PC3	C:\>ping 192.50.20.10	bytes=32 time=1ms TTL=124	bytes=32 time=1ms TTL=124	bytes=32 time=1ms TTL=124	bytes=32 time=10ms TTL=124	Sent=4, Received=4, Lost=0 (0% loss), Average=3ms